

PRATEESH PATLOLLA

Data Scientist / AI Engineer

(650) 466-8996 •

prateesh.reddy34@gmail.com •

prateeshreddy.github.io •

SUMMARY

Data Scientist with 5+ years of experience delivering Agentic AI, GenAI, NLP, Forecasting, and Optimization solutions at scale. Built cloud-native ML systems using Databricks and AWS, Azure, Spark, Databricks, and MLOps tools from experimentation to deployment. Delivered high-impact products including multi-agent architectures (LangGraph), Model Context Protocol (MCP) integrations, Bedrock, self-optimizing prompt loops, and Gurobi-based production optimizers. Skilled in A/B testing, Causal Inference, and cross-functional scenario modeling, with a track record of driving measurable ROI and operational efficiency across enterprise environments.

SKILLS

- **Languages:** Python, SQL, R, Java, JavaScript
- **Big Data & Pipelines:** Apache Spark, PySpark, Hadoop, Hive, Parquet, Kinesis Firehose, Glue
- **Cloud & MLOps:** Azure (Azure Databricks, Monitor, OpenAI), AWS (SageMaker, Bedrock, Redshift, Lambda, CloudWatch), Docker, Kubernetes, Airflow, Jenkins, FastAPI, GitHub Actions, VPC Networking
- **GenAI & LLMs:** LangChain, LangGraph, LangSmith Studio, Model Context Protocol (MCP), Multi-Agent Systems, LLM-as-a-judge, Guardrails AI, Apple Foundation Models (AFM), Llama 2, GPT-4, Vector DBs (FAISS, Pinecone, Chroma).
- **Databases & Storage:** Redis (ElastiCache), DynamoDB, PostgreSQL, Oracle, Redshift, MongoDB
- **Analytics & BI:** Tableau, Power BI, AWS QuickSight
- **Modeling & Optimization:** Scikit-learn, XGBoost, SARIMAX, PyTorch, SHAP, Gurobi, A/B Testing, Causal Inference
- **Monitoring:** Azure Monitor, CloudWatch, Datadog

PROFESSIONAL EXPERIENCE

Apple
Data Scientist

Austin, TX
Nov 2025 – present

- Architected a tool-calling Ops Agent deployed on LangGraph to streamline budget and roster planning for ~40 evaluation managers. Saving 100s of hours weekly by instantly executing complex scenario modeling such as calculating cost offsets for changing scope or reallocating feature budgets and autonomously pushing updates directly to Quip intake forms via Model Context Protocol (MCP).
- Engineered a self-improving optimization loop that automatically tunes system prompts and tool descriptions based on evaluation feedback and golden set data. Integrated strict observability, safety guardrails, and dynamic multi-model fallback chaining handling throttling.
- Developed a Guideline Alignment Quality Agent that continuously cross-references production system prompts against the latest evolving annotation guidelines, automatically identifying logic gaps and suggesting prompt patches to ensure completeness.
- Deployed LLM-as-a-judge Autograders to cut human annotators to evaluate real-world datasets while maintaining strict label fidelity.
- Developed a production anomaly detection ML model leveraging Apple Foundation Model (AFM) text embeddings to audit human evaluator workflows, accurately differentiating between AI-generated comments and human-written text to ensure QA dataset integrity and compliance.

Toyota North America
Data Scientist

Plano, TX
Feb 2023 – Nov 2025

- Led the design and POC of AskToyota, a GenAI chatbot using LangChain, FAISS, LangGraph, and LLaMA 2, deployed in secure VPC environments with IAM-based access control and TLS. Reduced stakeholder turnaround time by 30% by surfacing forecasts and KPIs on demand.
- Spearheaded Accessory Recommendation engine generating \$150M+ annual revenue using XGBoost, dynamic segmentation, and incentive-aware retraining, integrated A/B testing, and causal inference for robust evaluation.
- Built Toyota's Annual Production Optimizer (APO) using Gurobi to solve multi-objective LP problems (profit, volume, GHG score) across 14 global factories, optimizing \$700M+ in planning annually.
- Developed Smart Forecasting engine with PySpark and SARIMAX to predict dealer-level sales; enabled 3+ hours/month time savings per dealer and built self-serve analytics dashboards via Tableau.
- Productionized Pyspark and ML pipeline in Azure Databricks, integrating model registry, drift detection with auto-retraining, Airflow orchestration, and CloudWatch/Datadog monitoring for real-time alerts.
- Deployed real-time ETA API for dealer websites using ElastiCache Redis with SHA-1 for online inference; batch-trained models using PySpark and Parquet on S3. Packaged with Docker and deployed via Kubernetes.
- Blended causal inference and A/B testing frameworks with GenAI tools (Azure OpenAI, LangChain agents, Guardrails AI, Azure Cognitive Search, FAISS) on Azure Databricks to auto-explain accessory model KPIs, increasing dealer conversions by 20% and boosting stakeholder trust..

Amazon, Alexa AI
Data Scientist Intern

Santa Barbara, CA
May 2022 – August 2022

- Retrained and deployed Alexa's question categorizer using advanced NLP (multi-class classification, intent detection), improving query routing across 100+ categories via SageMaker.
- Used resampling and confidence intervals to evaluate and improve model confidence on rare query intents, increasing trust and accuracy for low-frequency categories in Alexa.
- Improved the generated success rate model's alignment with human-labeled success rate, reducing gap from ~20% to under 14%, enabling reduced annotation spend.
- Built reporting pipelines that pushed KPIs into Redshift and Quicksight to visualize key health metrics and rolling forecast 3-month trends to support weekly reviews with product stakeholders.

Cyient
Data Scientist

August 2019 – December 2020

- Extracted road sign boards through object detection from terrestrial imagery to minimize manual efforts of data annotation for North American-based clients.
- Achieved a hit rate of 92%, resulting in a saving of 12 FTEs.
- Designed and optimized AWS Glue ETL jobs to streamline data ingestion into a data warehouse, leading to significant improvements in data quality and processing speed.

EDUCATION

INDIANA UNIVERSITY BLOOMINGTON
Masters of Science in Data Science

Bloomington, Indiana, USA

GITAM UNIVERSITY
Bachelor's in Computer Science

Hyderabad, India

Prateesh's Gen AI Resume Assitant

Chatbot Link: prateeshreddy.github.io

- Developed a scalable Retrieval-Augmented Generation (RAG) pipeline using LangChain, FAISS, and OpenAI's embeddings, enabling context-aware, accurate Q&A about my professional background.
- Engineered advanced prompt templates and conversational memory, delivering enterprise-grade efficiency with 98% retrieval accuracy, 120 ms latency (p99), and inference costs below \$0.0005 per query.
- Deployed serverlessly on Cloudflare Workers with robust CI/CD via GitHub Actions, featuring real-time observability and automated logging for iterative improvements.

Cognitive Search Engine by NLP

- Provided Cognitive search capability for **Eli Lilly and Company** to search against databases like FDA and EMA via natural language questions and return relevant results to help with accelerating regulatory submissions for Eli Lilly as an Intern in the Summer of 2021 using Hugging Face transformers models.

Research Scientist at Institutional Analytics, Indiana University

- Been part of the guest lecture for ILSZ 637 - Information Visualization by Noriko Hara in Spring 2022, a graduate-level course talking about data storytelling and data visualization.
- Built data pipelines with Tableau reports to analyze student performance, retention, and graduation trends, influencing key academic decisions.